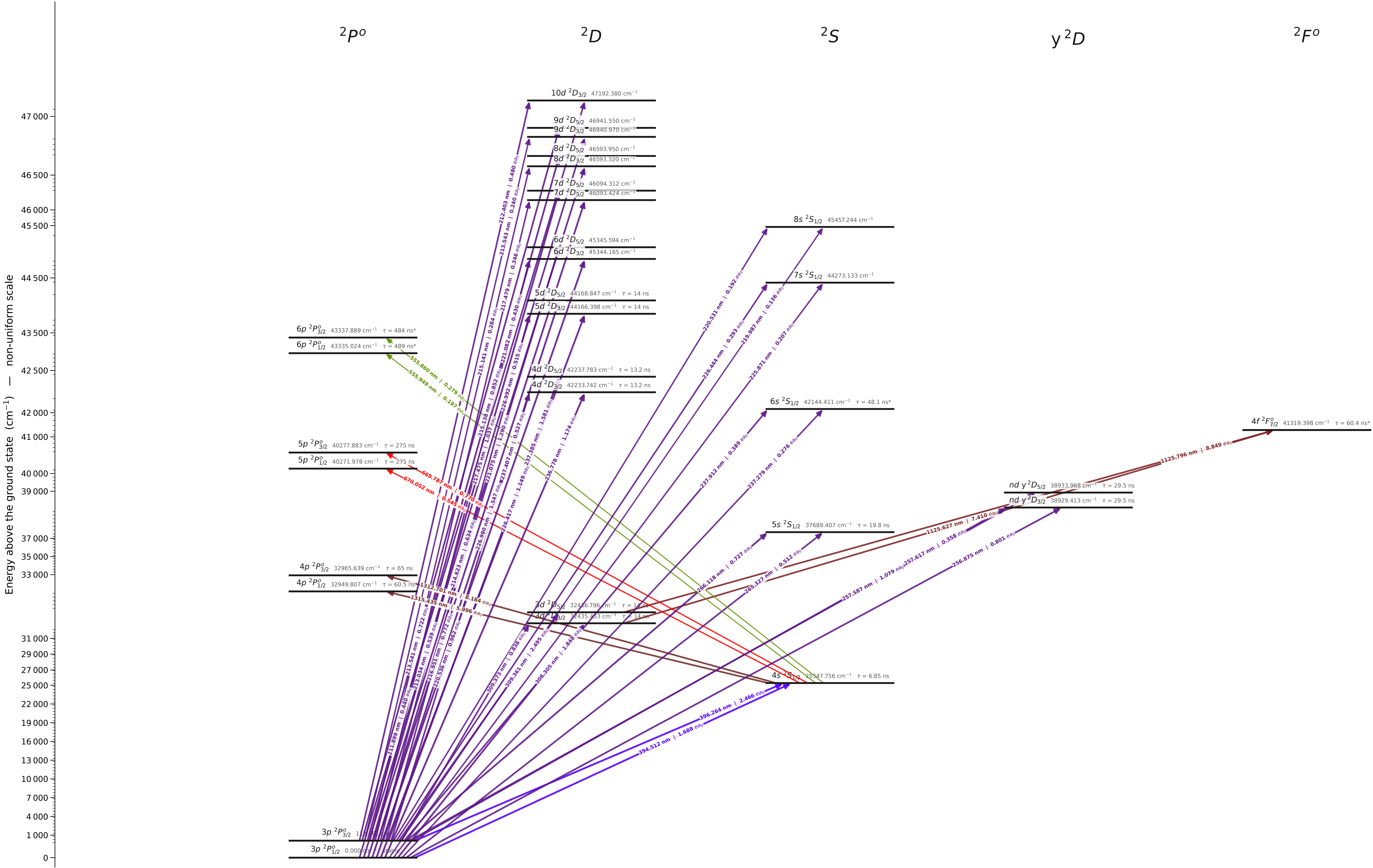
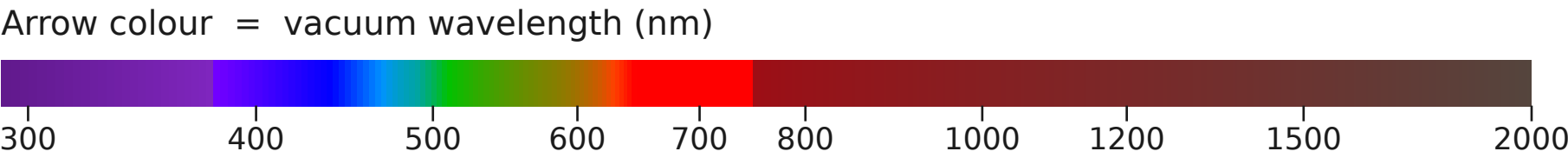


Aluminium

Al (Al I). Strongest classified transitions below 2 μm (NIST): 32 levels, 44 lines.

Al⁺ ionisation limit 48278.48 cm⁻¹ = 5.98577 eV (207.132 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ e r \ J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3p ² P _{3/2} - 3d ² D _{5/2}	309.3606	309.2708	969.0712	2.495	7.100e+07	11.37	99.4	309.4 nm laser-cooling transition 3p 2P3/2
3p ² P _{3/2} - 4s ² S _{1/2}	394.5122	394.4006	759.9066	1.688	4.700e+07	23.23	32.2	394.4 nm resonance line
3p ² P _{3/2} - 4s ² S _{1/2}	396.2641	396.1520	756.5471	2.466	9.900e+07	23.23	67.8	396.2 nm line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁷Al hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3p ² P _{3/2}	502.0346(5)	-	F=2: -878.56 F=3: +627.54
3p ² P _{3/2}	94.27723(10)	18.915	F=1: -481.71 F=2: -308.29 F=3: -33.97 F=4: +358.27
4s ² S _{1/2}	431.11(9)	-	F=2: -754.44 F=3: +538.89
4p ² P _{3/2}	58.280	-	F=2: -101.99 F=3: +72.85
4p ² P _{3/2}	23.120	2.860	F=1: -119.38 F=2: -75.43 F=3: -7.35 F=4: +87.42

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3p ² P _{3/2} - 4s ² S _{1/2}	396.264	26m-27	378(3)
3p ² P _{3/2} - 4s ² S _{1/2}	396.264	28-27	-321.9(8)
3p ² P _{3/2} - 4s ² S _{1/2}	396.264	29-27	-610.7(6)
3p ² P _{3/2} - 4s ² S _{1/2}	396.264	30-27	-889.5(12)
3p ² P _{3/2} - 4s ² S _{1/2}	396.264	31-27	-1141.0(12)
3p ² P _{3/2} - 4s ² S _{1/2}	396.264	32-27	-1401(7)

Shift = ν(first isotope) – ν(second isotope).